



MB 512: Mathematical Foundations for DS and AI

Course Code	MB 512
Course Name	Mathematical Foundations for DS and AI
L-T-P-C	2-0-0-2 (L-T-P-C)
Prerequisites	None
Intended for	MBA
Distribution	Compulsory
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Why mathematics for machine learning and artificial intelligence, concepts of models; constants, parameters and variables; mathematical models, simple and deterministic models. (2 hours)
- **Linear Algebra:** Linear equations and solutions, Scalars and Vectors, vector arithmetic and operations, orthogonality; Linear Independence, basis vectors; Matrices, basic matrix arithmetic and operations, rank of a matrix, matrix types, sparse matrix, matrix factorization, soft introduction to concepts of Tensors; Concepts of linearity and nonlinearity, linear Mappings, Vector and matrix norms; Eigenvectors and eigenvalues, singular value decomposition (SVD). (8 hours)
- **Calculus:** Limits and Functions, nature of Functions, univariate and multivariate functions, continuity of a function; basic functions like exponential, logarithmic, trigonometric, hyperbolic, modulus, greatest integer etc; squashing functions and activation functions; composite functions, Derivative, derivative of basic functions and activation functions, Chain rule, concepts of partial differentiation; Integrals, substitution rule, areas between curves. (6 hours)
- **Dimensions:** Concepts of dimensions-zero dimension to multiple dimensions, hyperspace, Euclidean space and dimensions, Euclidean distance between points, soft introduction to non-Euclidean space; vectors and matrices in dimensional space; Dimensions and analysis space; Business concepts and dimensions, mapping business problems into dimensional representation, multidimensional analysis. (4 hours)
- **Optimization:** Concepts of single and multiple attributes, objectives and criteria; Concepts of constraints and constraint satisfaction problems; Maximum and minimum of univariate and multivariate functions, Saddle Points, local and global optimum, concepts of linear and nonlinear optimization, constrained and unconstrained optimization, soft introduction to linear programming; search space, feasible and infeasible solution space, single agent and multi-agent search problems, search domain exploration and exploitation; Least squares method, Concept of gradient, gradient of vector valued functions, gradient of matrices; loss functions, gradient descent method. (8 hours)

Text Books

1. Nil

References

1. Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong, **Mathematics for Machine Learning**, Cambridge University Press, 2020 (<https://mml-book.com>)
2. Thomas Nield, **Essential Math for Data Science: Take Control of Your Data with Fundamental Linear Algebra, Probability, and Statistics**, Shroff Publishers & Distributors Pvt. Ltd., Mumbai, 2022.
3. Charu C. Aggarwal, **Linear Algebra and Optimization for Machine Learning**, Springer Nature, 2020.
4. MS Excel 2019, **Data Analysis & Business Modeling**, Wayne Winston, Microsoft Press (PHI), 2020.